

APPENDIX 5 - MEMBER CASE TABLES PER CONDITION CATEGORY (FROM CHAPTER 4)

Condition type: technology practices

Category 1: Technology Practices - Democratic Data Authority

Table 1 Member Cases for TP Category 1 - DDA

Case	Condition Name	Challenge Addressed
DECODE	Democratic Data Governance	Data controlled by corporations without citizen oversight
First Nations Information Governance Centre (FNIGC)	Democratic Data Governance	Historical governmental exclusion of First Nations from data collection
MAIAM NAYRI WINGARA	Democratic Data Governance	Mainstream data infrastructures framing Indigenous Australians as problems rather than agents
MiData	Democratic Data Governance	Health data controlled by corporations without individual consent
OpenSAFELY	Democratic Data Governance	Assumption that health research requires non-transparent data handling
Worker Algorithm Observatory (WAO)	Democratic Data Governance	Opaque algorithmic management hiding exploitation

Category 2: Technology Practices - Open Development and Transparency

Table 2 Member Cases for TP Category 2 - ODT

Case	Condition Name	Challenge Addressed
DECODE	Open-Source Development	Proprietary data systems limiting transparency and control
Linux Foundation	Open Access and Transparency	Proprietary software concentrating technological control
Linux Foundation	Open-Source Development	Single-entity control over critical software infrastructure
Loomio	Open-Source Development	Proprietary decision-making tools limiting organizational democracy
Matrix	Open-Source Development	Proprietary communication protocols limiting interoperability
OpenStreetMap	Open Access and Transparency	Proprietary maps limiting access and modification
Sensorica	Open-Source Development	Proprietary R&D excluding broader participation
WikiHouse	Open-Source Development	Proprietary construction systems limiting access to housing innovation

All rows address proprietary enclosure as the fundamental challenge. The mechanism is consistent: open processes that enable collective participation in development. Note that Linux Foundation has two rows in this category (reflecting two distinct conditions within the

same mechanism), and DECODE, Matrix, and OpenStreetMap appear in other categories as well (reflecting their multi-dimensional technology practices).

Category 3: Technology Practices - Multi-Scale Architecture Design

Table 3 Member Cases for TP Category 3 - MSAD

Case	Condition Name	Challenge Addressed
Australian Research Data Commons	Technology Design for Multi-Scale Governance	Single-scale infrastructure limiting research coordination
Guifi.net	Multi-Scalar Federated Governance	Tension between network interoperability and local community adaptation needs
Matrix	Technology Design for Multi-Scale Governance	Single-scale solutions failing to address varied governance needs
Open Mobility Foundation	Technology Design for Multi-Scale Governance	Proprietary mobility data controlled by private platforms
OpenAgriNet	Technology Design for Multi-Scale Governance	Predatory agricultural technology controlled by agribusiness
SOLshare	Technology Design for Multi-Scale Governance	Isolated solar systems unable to share excess capacity

All rows address scalar challenges: either single-scale limitations (ARDC, Matrix), isolation (SOLshare), or external control that multi-scale design can circumvent (Open Mobility Foundation, OpenAgriNet). The mechanism involves architectural choices that create affordances for both local governance and broader coordination. This directly connects to the dissertation's concern with how initiatives navigate tensions between local autonomy and trans-local engagement (RQ III).

Category 4: Technology Practices - Community-Centered Design Processes

Table 4 Member Cases for TP Category 4 - CCDP

Case	Condition Name	Challenge Addressed
DAIR (Distributed AI Research Institute)	Participatory Technology Design	Large-scale AI models exacerbating bias and inequality
The Drivers Cooperative	Transparent and Participatory Design	Tension between driver privacy concerns and cooperative transparency requirements

Both rows involve initiatives explicitly centering affected communities in design decisions. DAIR addresses exclusion from AI development; The Drivers Cooperative navigates internal tensions through participatory processes. The mechanism differs from

Democratic Data Authority (Category 1) in focus: Category 1 addresses governance of existing/collected data, while this category addresses the design process itself.

Category 5: Technology Practices - Culturally-Grounded Development

Table 5 Member Cases for TP Category 5 - CGD

Case	Condition Name	Challenge Addressed
Masakhane	Culturally-Embedded Technology Development	Tension between shared NLP frameworks and language-specific community needs

This case represents technology development that navigates the tension between using established technical frameworks and adapting to culturally-specific requirements. The mechanism involves design choices that prioritize cultural embeddedness over standardization. This connects to the dissertation's decolonial framing, challenging the assumption that technology should be culturally neutral.

Category 6: Technology Practices - Commons Resource Stewardship

Table 6 Member Cases for TP Category 6 - CRS

Case	Condition Name	Challenge Addressed
Digital Empowerment Foundation	Community-Owned Infrastructure Development	Rural internet infrastructure gaps and language barriers to government services
OpenStreetMap	Commons-Based Resource Governance	Tension between global data model consistency and local mapping priorities
Research Data Alliance	Open Standards Development	Tension between global interoperability requirements and disciplinary specificity

All rows involve stewardship of shared resources: infrastructure (DEF), mapping data (OpenStreetMap), or research data standards (RDA). The challenges addressed involve tensions between openness / standardization and community-specific needs. The mechanism differs from Category 2 (Open Development) in focus: Category 2 emphasizes the development process, while this category emphasizes the governance of resulting shared resources.

Condition type: resources

Category 1: Resources - Community Capacity Development

Table 7 Member Cases for R Category 1 - CCD

Case	Condition Name	Challenge Addressed
DAIR (Distributed AI Research Institute)	Community Capacity Building	Communities excluded from AI development processes
First Nations Information Governance Centre (FNIGC)	Community Capacity Building	First Nations lacking infrastructure for community-controlled data governance
Masakhane	Community Capacity Building	AI research capacity concentrated in Western institutions
Participatory Brazil	Community Capacity Building	Complexity of government processes limiting citizen engagement
Slum Dwellers International	Community Capacity Building	Slum communities lacking data about their own conditions
Stop LAPD Spying Coalition	Community Capacity Building	Communities unable to challenge surveillance systems

Category 2: Resources - Access and Skills Democratization

Table 8 Member Cases for R Category 2 - ASD

Case	Condition Name	Challenge Addressed
CLEAR (Civic Laboratory for Environmental Action Research)	Democratizing Access and Skills	Expensive specialized equipment excluding community participation
Detroit Community Technology Project (CTC)	Capacity Building and Technological Empowerment	Digital skills gap excluding residents from economic opportunities
Digital Empowerment Foundation	Democratizing Access and Skills	Digital exclusion limiting micro-entrepreneurs from accessing e-commerce
Zenzeleni	Capacity Building and Technological Empowerment	External technical dependence creating vulnerability

All cases address exclusion from participation due to skills gaps or equipment access barriers. Unlike Category 1 (which focuses on collective community capacity), this category emphasizes individual-level empowerment: giving people the tools and skills to participate. The challenges addressed span scientific equipment (CLEAR), digital skills (Detroit CTC, DEF), and technical self-reliance (Zenzeleni).

Category 3: Resources - Economic Value Redistribution

Table 9 Member Cases for R Category 3 - EVR

Case	Condition Name	Challenge Addressed
CoopCycle	Value Redistribution Mechanisms	Gig economy platforms transferring financial risks to workers while extracting commissions
Fairbnb.coop	Value Redistribution Mechanisms	Tourism platforms extracting value without community benefit
Guifi.net	Economic Transformation and Value Redistribution	Telecommunications revenues flowing to distant corporate shareholders
Namma Yatri	Economic Transformation and Value Redistribution	Corporate ride-hailing extracting 40% commissions from drivers
Solar Commons	Economic Transformation and Value Redistribution	Energy transition excluding low-income communities
The Drivers Cooperative	Economic Transformation and Value Redistribution	Corporate platforms monetizing driver data with opaque algorithms

All cases explicitly address extractive economic relations where value flows from communities / workers to external corporate actors. The mechanism is redistribution: changing the direction and destination of value flows. This directly connects to RQ II (community wealth) by demonstrating how initiatives restructure economic relations to generate community-level wealth rather than corporate profits.

Category 4: Resources - Alternative Economic Experimentation

Table 10 Member Cases for R Category 4 - AEE

Case	Condition Name	Challenge Addressed
Enspiral	Alternative Economic Models	Traditional funding models concentrating decision-making
Les Mercedes	Alternative Economic Models	Gig platforms externalizing risk through independent contractor classification
Sensorica	Alternative Economic Models	Traditional organizational forms limiting economic experimentation

All cases involve experimentation with economic models that differ from both traditional employment and platform capitalism. Unlike Category 3 (which redistributes value within existing economic logics), this category reimagines economic relationships entirely. Enspiral experiments with collective funding, Les Mercedes challenges contractor classification, Sensorica develops open value networks.

Category 5: Resources - Sustainable Resourcing Models

Table 11 Member Cases for R Category 5 - SRM

Case	Condition Name	Challenge Addressed
Australian Research Data Commons	Public-Community Co-Investment Models	Unsustainable reliance on single funding sources for essential research infrastructure
Linux Foundation	Economic Transformation	Financial dependence on single funding sources

Both cases address sustainability challenges facing commons-based initiatives, specifically the vulnerability that comes from depending on single funding sources. The mechanism involves diversifying resource streams to build organizational resilience. This differs from redistribution (Category 3) in focus: here the concern is organizational sustainability rather than community wealth capture.

Category 6: Resources - Values-Aligned Resource Allocation

Table 12 Member Cases for R Category 6 - VARA

Case	Condition Name	Challenge Addressed
Masakhane	Value-Based Resource Allocation	Lack of African language representation in AI technologies
OpenSAFELY	Bidirectional Knowledge Exchange	Resource scarcity in public health data research

Both cases involve directing resources toward underserved priorities: African languages in AI (Masakhane) and public health data research (OpenSAFELY). The mechanism differs from redistribution in that it focuses on what resources are directed toward (allocation) rather than who captures generated value (distribution).

Condition type: governance

Category 1: Governance - Decentralized Authority Distribution

Table 13 Member Cases for G Category 1 - DAD

Case	Condition Name	Challenge Addressed
DAIR (Distributed AI Research Institute)	Decentralized Decision-Making	Hierarchical control and concentrated decision-making in AI research
Detroit Community Technology Project (CTC)	Decentralized Decision-Making	Centralized top-down decision-making in technology projects
Enspirial	Decentralized Decision-Making	Hierarchical organizational control limiting participation

Loomio	Decentralized Decision-Making	Hierarchical organizational structures limiting democratic participation
Nea Guinea/Tzoumakers	Decentralized Decision-Making	Top-down technology development excluding local knowledge and participation
Participatory Brazil	Decentralized Decision-Making	Centralized state decision-making

Category 2: Governance - Ownership-Based Democratic Control

Table 14 Member Cases for G Category 2 - OBDC

Case	Condition Name	Challenge Addressed
CoopCycle	Worker Ownership Structures	Gig economy platforms extracting value from workers
Groupmuse	Community Ownership for Wealth Retention	Platform capitalism extracting value from cultural producers
MiData	Community Ownership for Wealth Retention	Commercial platforms monetizing personal health data
Som Energia	Community Ownership for Wealth Retention	Monopolistic structures in energy sector
The Drivers Cooperative	Worker Ownership Structures	Ride-hailing platforms taking 40% commissions leaving drivers with low wages
The Drivers Cooperative	Collective Ownership and Governance	Algorithmic management and unilateral corporate decision-making

All cases use ownership as the lever for governance change. Worker ownership (CoopCycle, The Drivers Cooperative) and community ownership (Groupmuse, MiData, Som Energia) both serve to embed democratic control in organizational structure. The challenges addressed are primarily extractive: ownership restructuring counters extraction by giving those who create value governance authority over it.

Category 3: Governance - Federated Multi-Scale Coordination

Table 15 Member Cases for G Category 3 - FMC

Case	Condition Name	Challenge Addressed
Guifi.net	Federated Governance Models	Centralized telecommunications control limiting community autonomy
Matrix	Federated Governance Models	Centralized communication platforms controlling user data
MiData	Federated Democratic Governance	Tension between local data sovereignty and global research coordination needs

All cases involve federated governance structures that enable multiple scales of authority to coexist. The challenges addressed include centralized control (Guifi.net, Matrix) and coordination needs (MiData). This category connects directly to RQ III concerning how initiatives navigate tensions between local autonomy and trans-local engagement; federation is a governance solution to scalar tensions.

Category 4: Governance - Democratic Scaling Strategies

Table 16 Member Cases for G Category 4 - DSS

Case	Condition Name	Challenge Addressed
CoopCycle	Scale and Democratic Participation Navigation	Tension between operational efficiency and maintaining democratic participation as organization scales
CoopCycle	Federated Scaling Strategy	Tension between federation growth and maintaining individual cooperative autonomy
First Nations Information Governance Centre (FNIGC)	Multi-Network Presence and Translocal Engagement	Tension between national focus and participation in global Indigenous sovereignty networks

All cases explicitly address scale-related governance challenges. CoopCycle navigates scaling within its federation; FNIGC navigates engagement across scales (national to global). The mechanism is intentional design to preserve democratic character during growth, recognizing that scale naturally creates pressures toward hierarchy and exclusion.

Category 5: Governance - Inclusive Expertise Integration

Table 17 Member Cases for G Category 5 - IEI

Case	Condition Name	Challenge Addressed
Sensorica	Stigmergic Coordination Systems	Hierarchical corporate R&D structures limiting participation in innovation
Worker Algorithm Observatory (WAO)	Inclusive Expertise Integration	Tension between broad worker participation and specialized analytical capacity

Both cases address the expertise-democracy tension: how can specialized knowledge be integrated with broad participation? Sensorica's stigmergic approach and WAO's inclusive expertise model represent different solutions. This is a distinct governance challenge from

decentralization (Category 1); the issue is not just distributing authority but integrating different types of knowledge.

Category 6: Governance - Democratization of External Processes

Table 18 Member Cases for G Category 6 - DEP

Case	Condition Name	Challenge Addressed
Participatory Brazil	Democratization of Governance and Technology	Top-down technocratic planning excluding citizen input
Slum Dwellers International	Democratization of Governance and Technology	Top-down urban planning ignoring resident knowledge

Both cases address exclusion from external governance processes (state planning in both cases). Unlike Category 1 (which addresses internal organizational decentralization), this category focuses on democratizing how communities engage with external institutions. The mechanism involves creating new participation channels in existing governance structures.

Condition type: technology, tools, data, products, and customers (22 rows of case information)

Category 1: Technology Tools - Community-Controlled Physical Infrastructure

Table 19 Member Cases for TT Category 1 - CPI

Case	Condition Name	Challenge Addressed
Detroit Community Technology Project (CTC)	Community-Owned Infrastructure	Lack of community control over digital communications infrastructure
Detroit Community Technology Project (CTC)	Community-Owned Infrastructure Development	Fixed telecommunications infrastructure limiting community autonomy
Guifi.net	Community-Owned Infrastructure	Lack of broadband infrastructure in rural areas where commercial ISPs found deployment unviable
SOLshare	Community-Owned Infrastructure	Millions lacking reliable electricity despite large off-grid solar programs
Telecomunicaciones Indígenas Comunitarias (TIC A.C.)	Community-Owned Infrastructure	Large commercial operators unwilling to serve indigenous communities
Zenzeleni	Community-Owned Infrastructure	Commercial telecommunications providers neglecting rural communities

Category 2: Technology Tools - Culturally-Grounded Technology Artifacts

Table 20 Member Cases for TT Category 2 - CTA

Case	Condition Name	Challenge Addressed
Digital Empowerment Foundation	Culturally-Embedded Technology Development	Language barriers accessing government welfare schemes
Monlam AI	Culturally-Embedded Technology Development	Technology development disconnected from Buddhist scholarly traditions
Nea Guinea/Tzoumakers	Culturally-Embedded Technology Development	Global design disconnect from local agricultural knowledge
Te Hiku Media	Culturally-Embedded Technology Development	Commercial AI inadequacy for Indigenous language processing

All cases involve technology development rooted in specific cultural contexts: Indian local languages (DEF), Tibetan Buddhist traditions (Monlam AI), Greek agricultural knowledge (Nea Guinea), Māori language (Te Hiku Media). The mechanism is embedding cultural specificity into technology rather than adapting communities to generic tools. This connects to decolonial framing, challenging the assumption that technology is or should be neutral.

Category 3: Technology Tools - Indigenous Data Sovereignty Infrastructure

Table 21 Member Cases for TT Category 3 - IDSI

Case	Condition Name	Challenge Addressed
First Nations Information Governance Centre (FNIGC)	Indigenous Data Sovereignty	Lack of First Nations tools for community-controlled data governance
Global Indigenous Data Alliance (GIDA)	Indigenous Data Sovereignty	Conventional metadata systems ignoring Indigenous governance protocols

Both cases address the gap between Indigenous data sovereignty as principle and as operational capacity. The challenges involve lack of appropriate tools (FNIGC) and inappropriate existing systems (GIDA). The mechanism is developing purpose-built infrastructure that embeds Indigenous governance protocols. This directly operationalizes sovereignty-as-capacity: sovereignty requires actual technical infrastructure, not just declarations.

Category 4: Technology Tools - Participatory Design Platforms

Table 22 Member Cases for TT Category 4 - PDP

Case	Condition Name	Challenge Addressed
Loomio	Transparent and Participatory Design	Exclusion from decision-making due to scheduling or location barriers
Open Mobility Foundation	Transparent and Participatory Design	Opaque technical decision-making in mobility infrastructure
Participatory Brazil	Transparent and Participatory Design	Limited citizen participation in government decision-making

All cases involve platforms designed to enable participation. The challenges address different barriers: logistical (Loomio), technical opacity (OMF), and institutional exclusion (Participatory Brazil). The mechanism is technology design that creates affordances for democratic participation; the platform itself embodies participatory values.

Category 5: Technology Tools - Digital Commons Resources

Table 23 Member Cases for TT Category 5 - DCR

Case	Condition Name	Challenge Addressed
OpenStreetMap	Digital Resources as Commons	Proprietary mapping controlled by corporate interests
OpenStreetMap	Open Source Commons Development	High operational costs limiting mapping accessibility
P2P Foundation	Digital Resources as Commons	Enclosure of knowledge and technology through intellectual property
WikiHouse	Digital Resources as Commons	Construction knowledge enclosed within proprietary systems

All cases involve treating digital resources as commons. The challenges address enclosure through proprietary control (OpenStreetMap, P2P, WikiHouse) and accessibility barriers (OpenStreetMap). The mechanism is commons governance; neither purely open nor privately controlled, but collectively stewarded. OpenStreetMap has two rows reflecting both the commons nature and the accessibility dimension.

Category 6: Technology Tools - Algorithmic Transparency Infrastructure

Table 24 Member Cases for TT Category 6 - ATI

Case	Condition Name	Challenge Addressed
Stop LAPD Spying Coalition	Algorithmic Accountability and Transparency	Hidden institutional networks sustaining surveillance technologies

Worker Algorithm Observatory (WAO)	Algorithmic Accountability and Transparency	Worker exclusion from algorithmic management investigations
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Both cases involve creating tools for algorithmic accountability. The challenges address opacity in surveillance systems (Stop LAPD) and workplace algorithmic management (WAO). The mechanism differs from other categories: rather than building alternative infrastructure, these initiatives build infrastructure for understanding existing algorithmic systems. This represents a distinct pathway: transparency and accountability instead of replacement.

Category 7: Technology Tools - Decentralized Technical Architecture

Table 25 Member Cases for TT Category 7 - DTA

Case	Condition Name	Challenge Addressed
SOLshare	Decentralized Technological Architecture	Centralized energy utility control and lack of peer-to-peer energy trading

This case represents decentralized technical architecture as a distinct approach, enabling peer-to-peer energy trading through technical design. The mechanism is architectural: the technology itself is designed to enable decentralized coordination. This differs from Category 1 (community ownership of infrastructure) in focus. Here the emphasis is on the architectural design choice rather than ownership structure.

Condition type: partners

Category 1: Partners - Reciprocal Knowledge Exchange

Table 26 Member Cases for P Category 1 - RKE

Case	Condition Name	Challenge Addressed
Australian Research Data Commons	Bidirectional Knowledge Exchange	Research data fragmented across institutions
DECODE	Bidirectional Knowledge Exchange	Lack of citizen control over data governance
Digital Empowerment Foundation	Bidirectional Knowledge Exchange	Unidirectional knowledge transfer from global to local
Masakhane	Bidirectional Knowledge Exchange	Academic research disconnected from local linguistic communities

P2P Foundation	Bidirectional Knowledge Exchange	Unidirectional knowledge transfer from centers to peripheries
Research Data Alliance	Bidirectional Knowledge Exchange	Fragmented data sharing limiting research collaboration

Category 2: Partners - Multi-Network Strategic Positioning

Table 27 Member Cases for P Category 2 - MSP

Case	Condition Name	Challenge Addressed
Enspiral	Multi-Network Presence	Isolated social enterprises lacking mutual support
Global Indigenous Data Alliance (GIDA)	Multi-Network Presence	Fragmented Indigenous data sovereignty movements
Open Mobility Foundation	Multi-Network Presence	Cities isolated in negotiations with mobility platforms
P2P Foundation	Multi-Network Presence	Isolated commons initiatives lacking coordination
Slum Dwellers International	Multi-Network Presence	Isolated slum communities lacking collective voice
Te Mana Raraunga	Multi-Network Presence	Isolated Indigenous movements lacking collective voice

All cases address isolation as the fundamental challenge. The mechanism is network-building that enables collective action across fragmented actors. The contexts vary across social enterprises (Enspiral), Indigenous movements (GIDA, Te Mana Raraunga), cities (OMF), commons initiatives (P2P), and slum communities (SDI), but the logic is consistent: isolation weakens, networks strengthen.

Category 3: Partners - Community Sovereignty Protection in Partnerships

Table 28 Member Cases for P Category 2 - CSPP

Case	Condition Name	Challenge Addressed
Fairbnb.coop	Community Sovereignty Within Global Networks	Tension between local community accountability and platform expansion
Monlam AI	Community-Controlled Technology Development	Tension between external technical partnerships and monastic/scholarly oversight
Te Hiku Media	Indigenous Data Sovereignty	Tension between potential corporate partnerships and maintaining Māori control over language data

All cases explicitly address tensions between partnership opportunities and sovereignty maintenance. The challenges involve platform expansion (Fairbnb), external

technical partnerships (Monlam AI), and corporate partnerships (Te Hiku Media). The mechanism is navigating these tensions through governance arrangements that protect community control while enabling beneficial collaboration.

Category 4: Partners - Expertise-Community Knowledge Integration

Table 29 Member Cases for P Category 4 - EKI

Case	Condition Name	Challenge Addressed
CLEAR (Civic Laboratory for Environmental Action Research)	Decolonial Research Ethics	Tension between scientific expertise and community knowledge authority
Linux Foundation	Open Source Commons Development	Tension between corporate funding contributions and community-led governance
Masakhane	Inclusive Expertise Integration	Tension between technical AI specialization and broad linguistic community participation
OpenSAFELY	Privacy-Preserving Technology Design	Tension between researcher access needs and patient data protection

All address expertise-community tensions in partnerships. The challenges involve scientific vs. community knowledge (CLEAR), corporate vs. community governance (Linux Foundation), technical vs. linguistic expertise (Masakhane), and researcher vs. patient interests (OpenSAFELY). The mechanism is developing hybrid governance that integrates rather than subordinates different knowledge forms.

Category 5: Partners - Cross-Sector Collaboration Networks

Table 30 Member Cases for P Category 5 - CCN

Case	Condition Name	Challenge Addressed
Sensorica	Multi-Sector Innovation Networks	Closed innovation silos limiting collaboration between corporate, academic and community actors

Sensorica (only case) represents a distinct approach, bridging sector boundaries rather than building within-sector networks (Category 2) or managing bilateral partnerships (Categories 3, 4). The mechanism is creating infrastructure for cross-sector collaboration that traditional institutional boundaries prevent.

Condition type: purpose (16 rows of case information)

Category 1: Purpose - Indigenous Epistemological Centering

Table 31 Member Cases for Pu Category 1 - IEC

Case	Condition Name	Challenge Addressed
First Nations Information Governance Centre (FNIGC)	Indigenous Knowledge Systems Integration	Colonial data practices ignoring Indigenous knowledge protocols
Global Indigenous Data Alliance (GIDA)	Indigenous Knowledge Systems Integration	Open data frameworks ignoring Indigenous sovereignty
Te Hiku Media	Indigenous Knowledge Systems Integration	Western governance models disconnected from Māori values
Te Mana Raraunga	Indigenous Knowledge Systems Integration	Current data governance structures fail to reflect Māori ways of knowing

Category 2: Purpose - Cultural Preservation and Linguistic Sovereignty

Table 32 Member Cases for Pu Category 2 - CPLS

Case	Condition Name	Challenge Addressed
Masakhane	Cultural Sovereignty and Knowledge Preservation	Exclusion of African languages from digital technologies
Monlam AI	Cultural Sovereignty and Knowledge Preservation	Urgent threat to Tibetan language and cultural heritage
Te Hiku Media	Cultural Sovereignty and Knowledge Preservation	Most AI language models fail to process Indigenous languages

All three cases frame technology work as cultural/linguistic preservation. The challenges are existential; language exclusion threatens cultural survival. The mechanism is purpose orientation: technology development serves cultural preservation rather than commercial or efficiency goals. This represents a distinct framing from Category 1, which centers Indigenous epistemology in governance; Category 2 orients toward cultural preservation as mission.

Category 3: Purpose - Value System Reconceptualization

Table 33 Member Cases for Pu Category 3 - VSR

Case	Condition Name	Challenge Addressed
Groupmuse	Value Reconceptualization	Commercial music platforms prioritizing profit over cultural value
MAIAM NAYRI WINGARA	Value Reconceptualization	Colonial frameworks treating Indigenous data as public property

Solar Commons	Community Asset Development Through Legal Innovation	Tension between economic sustainability metrics and intangible community empowerment outcomes
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These cases challenge dominant value frameworks. Groupmuse challenges profit over culture; MAIAM NAYRI WINGARA challenges data-as-property; Solar Commons challenges narrow sustainability metrics. The mechanism is reconceptualization, or changing what counts as valuable. This directly connects to RQ II on community wealth by suggesting wealth must be redefined beyond monetary measures.

Category 4: Purpose - Resistance-Construction Integration

Table 34 Member Cases for Pu Category 4 - RCI

Case	Condition Name	Challenge Addressed
Telecomunicaciones Indígenas Comunitarias (TIC A.C.)	Resistance and Construction Integration	Regulatory burdens designed for commercial operators threatening community networks
Worker Algorithm Observatory (WAO)	Resistance and Construction Integration	Workers unable to understand or challenge algorithmic management

Both cases explicitly integrate resistance and construction. TIC A.C. resists regulatory burdens while building community networks; WAO resists algorithmic opacity while building worker knowledge. The mechanism is unified purpose that refuses separation between critique and creation. This represents a distinct political orientation.

Category 5: Purpose - Ecological and Social Transformation

Table 35 Member Cases for Pu Category 5 - EST

Case	Condition Name	Challenge Addressed
Les Mercedes	Feminist Economic Transformation	Patriarchal exclusion of women and marginalized workers from high-mobility urban labor sectors
Som Energia	Community Resilience and Ecological Sustainability	Tension between quantifiable energy metrics and broader social transformation measurement

Both embed technology / organizational work within broader transformation agendas: feminist transformation (Les Mercedes) and ecological sustainability (Som Energia). The

mechanism is purpose integration, where technology serves broader social / ecological transformation rather than being an end in itself.

Category 6: Purpose - Democratic Citizenship Obligation

Table 36 Member Cases for Pu Category 6 - DCO

Case	Condition Name	Challenge Addressed
Participatory Brazil	Purpose as Civic Obligation	Democratic deficits requiring active citizen participation as responsibility rather than optional engagement
Telecomunicaciones Indígenas Comunitarias (TIC A.C.)	Prioritizing Local Needs in Global Engagement	External actors imposing technology without community input

Both cases frame purpose in terms of citizenship and local priority. Participatory Brazil frames participation as obligation; TIC A.C. prioritizes local needs over external agendas. The mechanism is purpose orientation toward democratic citizenship and community priority rather than service provision or market competition.

Condition type: organizational information

Category 1: Org Info - Worker/Community Ownership Models

Table 37 Member Cases for OI Category 1 - WOM

Case	Condition Name	Challenge Addressed
Loomio	Collective Ownership and Governance	Hierarchical organizational structures limiting democratic software development
MiData	Cooperative Data Stewardship	Corporate control over personal health data without democratic governance
Namma Yatri	Worker-Owned Platform Alternative	Corporate platform extraction and algorithmic exploitation of gig workers

Category 2: Org Info - Decentralized Governance Architectures

Table 38 Member Cases for OI Category 2 - DGA

Case	Condition Name	Challenge Addressed
Guifi.net	Decentralized Decision-Making	Centralized control by telecommunications corporations
Participatory Brazil	Hybrid Democratic Participation Architecture	Limited citizen participation in government policy development

Both organizations are defined through governance structure. Guifi.net is decentralized; Participatory Brazil uses hybrid architecture. The organization IS its decision-making structure.

Category 3: Org Info - Anti-Extractive Value Circulation

Table 39 Member Cases for OI Category 3 - AVC

Case	Condition Name	Challenge Addressed
Fairbnb.coop	Anti-Extractive Value Distribution	Corporate platform extraction without community benefit
Guifi.net	Community-Controlled Economic Circulation	Corporate telecommunications extraction from local economies
Zenzeleni	Community Revenue Retention	Corporate value extraction from local communities

All three define organizations through value flow patterns. The challenges all address corporate extraction; the organizational identity is structured around preventing extraction. The mechanism is identity through economic relations; the organization IS its value circulation model.

Category 4: Org Info - Commons-Based Production Models

Table 40 Member Cases for OI Category 4 - CBPM

Case	Condition Name	Challenge Addressed
Matrix	Open Standards Development	Proprietary communication protocols fragmenting secure messaging
P2P Foundation	Commons-Based Knowledge Production	Knowledge fragmentation limiting commons movement coordination
WikiHouse	Open Source Manufacturing Networks	Proprietary construction systems limiting housing innovation access

All three define organizations through production mode: open standards (Matrix), commons-based knowledge (P2P), open source manufacturing (WikiHouse). The mechanism is identity through how outputs are created and governed.

Category 5: Org Info - Local-Global Scalar Positioning

Table 41 Member Cases for OI Category 6 - LGSP

Case	Condition Name	Challenge Addressed
Digital Empowerment Foundation	Community Capacity Building	Digital divides and technological dependency

Les Mercedes	Prioritizing Local Needs in Global Engagement	Growing precariousness of platform capitalism for delivery workers
Slum Dwellers International	Prioritizing Local Needs in Global Engagement	Top-down urban planning ignoring slum resident priorities

All cases define organizations through scalar positioning. DEF addresses local capacity; Les Mercedes and SDI prioritize local needs within global engagement. The organization IS its approach to local-global relations.

Category 6: Org Info - Municipal/Public Digital Governance

Table 42 Member Cases for OI Category 6 - MPDG

Case	Condition Name	Challenge Addressed
Open Mobility Foundation	Municipal Digital Infrastructure Governance	Corporate control over urban mobility data and lack of public oversight

OMF is defined through its municipal governance role. The mechanism is identity through public accountability; the organization IS its public governance function.

Condition type: politics (13 rows of case information)

Category 1: Politics - Policy and Systemic Transformation Advocacy

Table 43 Member Cases for Po Category 1 - PSTA

Case	Condition Name	Challenge Addressed
CLEAR (Civic Laboratory for Environmental Action Research)	Policy Influence and Systemic Change	Environmental governance excluding community scientific input
First Nations Information Governance Centre (FNIGC)	Policy Influence and Systemic Change	Regulatory frameworks marginalizing Indigenous data governance
Global Indigenous Data Alliance (GIDA)	Policy Influence and Systemic Change	International data standards ignoring Indigenous governance principles
Matrix	Policy Influence and Systemic Change	Proprietary protocols fragmenting secure communications
Open Mobility Foundation	Policy Influence and Systemic Change	Private platforms controlling public mobility data
P2P Foundation	Policy Influence and Systemic Change	Policy discourse ignoring commons-based alternatives
Stop LAPD Spying Coalition	Policy Influence and Systemic Change	Legal framework justifying surveillance expansion
Te Mana Raraunga	Policy Influence and Systemic Change	National data collection excluding Māori governance
The Drivers Cooperative	Policy Influence and Systemic Change	Policy acceptance of exploitative platform business models

Category 2: Politics - Resistance-Opposition Integration

Table 44 Member Cases for Po Category 2 - ROI

Case	Condition Name	Challenge Addressed
CLEAR (Civic Laboratory for Environmental Action Research)	Resistance and Construction Integration	Scientific expertise divorced from community priorities
Stop LAPD Spying Coalition	Resistance and Construction Integration	Police surveillance technologies preemptively criminalizing communities

Both initiatives explicitly integrate resistance and construction. CLEAR resists expert-community disconnection; Stop LAPD resists surveillance expansion. The mechanism is political engagement that refuses to separate opposition from alternative-building. This differs from Category 1's policy influence in its explicitly oppositional stance.

Category 3: Politics - Strategic Institutional Navigation

Table 45 Member Cases for Po Category 3 - SIN

Case	Condition Name	Challenge Addressed
First Nations Information Governance Centre (FNIGC)	Strategic Institutional Engagement	Colonial data practices failing to recognize Indigenous sovereignty
Namma Yatri	Worker-Owned Platform Alternative	Tension between market service provision and transformative non-extractive values

Both cases involve navigating institutional contexts that are not fully supportive. FNIGC engages colonial institutions strategically; Namma Yatri operates in markets while maintaining non-extractive values. The mechanism is strategic positioning that enables survival and impact within hostile or constraining systems.

Condition type: regulation

Category 1: Regulation - Legal Innovation for Sovereignty Protection

Table 46 Member Cases for Re Category 1 - LISP

Case	Condition Name	Challenge Addressed
CoopCycle	Legal Innovation for Community Control	Corporate appropriation of open-source cooperative technology
Guifi.net	Legal Innovation for Community Control	Infrastructure privatization threatening community networks
Solar Commons	Legal Innovation for Community Control	Renewable energy benefits flowing to external investors

Te Hiku Media	Legal Innovation for Community Control	Open licensing enabling appropriation of Indigenous language data
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Category 2: Regulation - Ethical Governance Framework Development

Table 47 Member Cases for Re Category 2 - EGFD

Case	Condition Name	Challenge Addressed
Australian Research Data Commons	Ethical Data Governance Frameworks	Tension between research data openness and protection of sensitive community information
OpenSAFELY	Ethical Data Governance Frameworks	Tension between international health data standards and NHS-specific governance requirements

Both cases address data governance tensions through ethical framework development. ARDC navigates openness vs. protection; OpenSAFELY navigates international vs. local standards. The mechanism is principled framework development to create replicable governance approaches.

Category 3: Regulation - Strategic Regulatory Navigation

Table 48 Member Cases for Re Category 3 - SRN

Case	Condition Name	Challenge Addressed
First Nations Information Governance Centre (FNIGC)	Strategic Navigation of Regulatory Landscapes	Hostile regulatory environments designed to privilege state control over Indigenous sovereignty
Open Mobility Foundation	Strategic Regulatory Navigation	Private mobility platforms controlling public infrastructure data
Telecomunicaciones Indígenas Comunitarias (TIC A.C.)	Strategic Regulatory Navigation	Mexican regulatory authority imposing \$50,000 annual spectrum fees

All examples involve navigating regulatory systems not designed for community initiatives. FNIGC navigates colonial regulatory environments; OMF navigates platform-dominated contexts; TIC A.C. navigates commercial-scale fee structures. The mechanism is strategic adaptation to survive within inappropriate frameworks.

Category 4: Regulation - Self-Determined Sovereignty Assertion

Table 49 Member Cases for Re Category 4 - SSA

Case	Condition Name	Challenge Addressed
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Global Indigenous Data Alliance (GIDA)	Sovereignty Reimagined Through Digital Self-Determination	State-centric legal frameworks that fail to recognize Indigenous data sovereignty
OpenAgriNet	Decentralized Technological Architecture	Tension between global agricultural data standards and smallholder farmer governance needs

Both involve asserting governance authority beyond state frameworks. GIDA reimagines sovereignty outside state recognition; OpenAgriNet addresses farmer governance needs that global standards ignore. The mechanism is self-determination.

Condition type: social context

Category 1: Social Context - Inclusive Participation Structures

Table 50 Member Cases for SC Category 1 - IPS

Case	Condition Name	Challenge Addressed
CLEAR (Civic Laboratory for Environmental Action Research)	Inclusive Participation and Representation	Western academic knowledge hierarchies marginalizing diverse perspectives
Digital Empowerment Foundation	Inclusive Participation and Representation	Systemic marginalization from decision-making processes
Participatory Brazil	Inclusive Participation and Representation	Government policy-making excluding citizen perspectives
Stop LAPD Spying Coalition	Inclusive Participation and Representation	Surveillance activism led by those distant from impacts

Category 2: Social Context - Movement and Network Embeddedness

Table 51 Member Cases for SC Category 2 - MNE

Case	Condition Name	Challenge Addressed
Global Indigenous Data Alliance (GIDA)	Embeddedness in Social Movements	Isolation of local Indigenous communities in challenging colonial data governance systems
OpenStreetMap	Global-Local Network Formation	Tension between global technical standards and local community mapping needs

Both cases involve embedding initiatives within broader movements or networks. GIDA connects to Indigenous sovereignty movements; OpenStreetMap navigates global-local network dynamics. The mechanism is movement connection that overcomes isolation and enables collective action.

Category 3: Social Context - Climate Crisis and Ecological Response

Table 52 Member Cases for SC Category 3 - CCEP

Case	Condition Name	Challenge Addressed
Slum Dwellers International	Climate Justice Integration	Climate vulnerability compounded by informal settlement marginalization and lack of adaptive infrastructure
SOLshare	Climate Crisis Response	Climate vulnerability compounded by centralized energy infrastructure failures

Both examples address climate vulnerability compounded by structural marginalization. SDI addresses informal settlement vulnerability; SOLshare addresses energy infrastructure failures. The mechanism is technology development that responds to climate crisis while addressing underlying inequalities.

Condition type: problem (8 rows of case information)

Category 1: Problem - Infrastructure and Access Deficits

Table 53 Member Cases for Pr Category 1 - IAD

Case	Condition Name	Challenge Addressed
Guifi.net	Community-Owned Infrastructure Development	Lack of broadband infrastructure in rural areas where commercial ISPs found deployment unviable
Linux Foundation	Open Source Commons Development	Proprietary software control and closed development processes

Category 2: Problem - Colonial and Western-Centric Knowledge Domination

Table 54 Member Cases for Pr Category 2 - CWKD

Case	Condition Name	Challenge Addressed
Masakhane	Decolonial Technology Development	Western dominance in AI development excluding African languages and perspectives
Te Mana Raraunga	Decolonial Research Ethics	Western-centric data governance excluding Indigenous epistemologies

Both examples identify Western / colonial dominance as the core problem. Masakhane addresses AI development; Te Mana Raraunga addresses data governance. The mechanism of response is decolonial approaches that center marginalized epistemologies.

Category 3: Problem - Cultural Erasure and Heritage Threat

Table 55 Member Cases for Pr Category 3 - CEHT

Case	Condition Name	Challenge Addressed
Monlam AI	Cultural Heritage Preservation through Technology	Political suppression of Tibetan language and cultural erasure through technological exclusion
Te Hiku Media	Alternative Value Definition	Narrow economic definitions of wealth excluding cultural and social dimensions

Both cases identify cultural / linguistic erasure as the core problem. Monlam AI addresses political suppression and technological exclusion; Te Hiku Media addresses economic value frameworks that ignore cultural dimensions. The mechanism of response is technology development oriented toward cultural preservation.

Category 4: Problem - Algorithmic Opacity and Asymmetric Power

Table 56 Member Cases for Pr Category 4 - AOAP

Case	Condition Name	Challenge Addressed
Matrix	Technological Sovereignty Architecture	Tension between local control and global interoperability in communications infrastructure
Worker Algorithm Observatory (WAO)	Worker-Led Transparency Mechanisms	Information asymmetry and algorithmic opacity in platform work

Both problems address opacity and asymmetric power in technological systems. Matrix addresses communications infrastructure sovereignty; WAO addresses algorithmic management opacity. The mechanism of response is transparency and community / worker-led accountability.